

the explosion of viruses being created by new authors and virus computer aided design and manufacturing tools (V-CAD/CAM).

It has been proposed that audit tools such as COPS can be used to deter virus infections because they strengthen the system's ability to control access and data movement. These inspection tools only improve control. As stated, control does not provide protection against virus attacks. It attempts to keep outside people out and inside people within their areas of authorization.

The third category of protection is Integrity. Integrity systems are intended to detect change. In the MS-DOS world, early integrity systems used cyclic redundancy character, CRC, values to detect change. A virus was then created which countered this protection. The virus determined the CRC value of the target file, infected it, and then padded the file until the CRC value computed the same. Many Unix users still use this method of change detection, or worse, they attempt to use the date of last modification as an indication of change. The date of last modification can be changed to any value on Unix systems with a simple user command. On many systems an option of the "touch" command provides this ability.

Any integrity tool that does not use cryptographic methods is of little value. In fact, if the integrity system fails to detect critical changes, then the false sense of security created in the system operator can be devastating to the system. An integrity tool, CIT, was created using the RSA Associates MD5 cryptographic hash algorithm. Since the algorithm is cryptographic, it can detect even a single bit flip and cannot be misled by any known means. In addition, during the development of CIT, it was determined that it was necessary to detect additions and deletions to the file system since these could be indications of non-infectious attacks such as performed by Trojan horses, worms and hackers. In this way, a rolling baseline can be created that will allow the system operator to quickly recover from any form of file system attack. Modifications to the protected